Newcastle University

School of Computing

MSc Data Science

CSC8639 — Dissertation

An AI-Enabled Digital Twin for EV Charging:

Behavioural Simulation, Policy Evaluation,

and Pricing Optimisation

Interim Report

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1 Introduction

The rapid global adoption of electric vehicles (EVs) represents one of the most consequential transitions in modern transportation infrastructure. The International Energy Agency reported that global EV stock surpassed 40 million vehicles in 2023, with projections indicating EVs could constitute over one-third of new vehicle sales by 2030 [1]. In the United Kingdom, the government's Net Zero Strategy mandates the phase-out of new petrol and diesel vehicle sales by 2035, placing EV charging infrastructure at the centre of national decarbonisation policy [2].

The growth of EV adoption introduces significant operational challenges for charging networks. Demand is inherently heterogeneous, shaped by driver behaviour, time-of-day patterns, connector type preferences, and real-time grid carbon intensity. Static or rule-based approaches to charging management are poorly suited to this complexity, as they cannot adapt to dynamic demand, varying price signals, or grid constraints [7].

Digital twins — computational replicas of physical systems calibrated to reproduce real-world behaviour — have emerged as a promising paradigm for addressing these challenges. When implemented as agent-based simulations and calibrated on real historical data, digital twins allow policymakers to test interventions before deployment, without disrupting live infrastructure [4]. Agent-based modelling (ABM) is particularly well-suited to EV charging contexts, as heterogeneous driver agents can make autonomous decisions based on energy need, cost sensitivity, and station availability [3, 10].

Despite this potential, few studies have grounded digital twin simulations in real institutional charging data, and fewer still have applied reinforcement learning (RL) to autonomously discover optimal pricing policies [12, 14]. This project addresses both gaps by building a fully calibrated, AI-enabled digital twin of the EV charging network at Newcastle University's Urban Sciences Building (USB), using 29,775 real historical charging sessions spanning 2021 to 2024.

2 Aim and Objectives

2.1 Aim

This project aims to develop, calibrate, and evaluate an AI-enabled agent-based digital twin of the EV charging network at Newcastle University's Urban Sciences Building, using real historical session data to simulate heterogeneous charging behaviour, evaluate policy interventions under realistic uncertainty, and optimise pricing and incentive strategies for grid resilience and carbon reduction.

2.2 Objectives

O1. To construct a reproducible data engineering pipeline that extracts, validates, and stores historical EV charging session data from the USB charging network — comprising 29,775 sessions across six chargers from March 2021 to July 2024 — in a portable analytical database suitable for simulation calibration and downstream analysis.

O2. To conduct a comprehensive exploratory data analysis of the charging session dataset, fitting statistical distributions to empirical charging behaviour including hourly arrival patterns, energy demand, session duration, and connector type preferences, and storing calibration parameters for use in the agent-based simulation.

O3. To implement a calibrated agent-based simulation using the Mesa framework, in which heterogeneous EV driver agents reproduce key statistical properties of the historical dataset — including arrival distributions, energy demand, session duration, and charger utilisation — within a defined tolerance threshold.

O4. To develop a scenario engine that stress-tests three policy interventions against the baseline simulation: EV adoption growth via scaled arrival rates; time-of-use (ToU) pricing via hour-varying price signals; and carbon-aware incentives via penalties on high-carbon charging periods — evaluating outcomes across four KPIs: grid load, station utilisation, average waiting time, and CO_2 emissions.

O5. To produce a quantitative policy evaluation report comparing baseline and intervention scenarios across all KPIs, with statistical analysis across multiple simulation runs to account for stochastic variability.

O6. *(Extension)* To implement a reinforcement learning policy optimisation module using Proximal Policy Optimisation (PPO) [11] via Stable-Baselines3 [9], wrapping the Mesa simulation as a Gymnasium-compatible environment [13], and demonstrating that the learned pricing policy outperforms handcrafted rules across the defined KPI set.

3 Overview of Progress

3.1 Literature Review

A targeted review of five papers has been conducted to ground the project methodology in current research. Papers span multi-agent EV charging simulation, digital twin frameworks for EV infrastructure, grid-integrated demand response, smart meter analytics for building energy benchmarking, and edge-AI reinforcement learning optimisation. Table 1 summarises the key contributions and their relevance.

Table 1: Summary of reviewed literature and relevance to this project.

Paper	Key Contribution	Relevance
Multi-Agent Game-Theoretic Model [3]	Utility-based EV agent with SOC urgency; dynamic pricing reduces peak congestion	Informs agent decision model and ToU pricing scenario
Digital Twin for EV Infrastructure [4]	ABM with PSO optimisation and GIS integration on a campus network	Validates Mesa as simulation framework; scenario engine design
Grid-Integrated DT Framework [5]	CPO incentive optimisation and demand response KPIs	Informs KPI selection: grid load, waiting time, CO_2
Smart City DT for Building Energy [6]	Smart meter analytics and temporally-segmented benchmarking	Informs data pipeline design and metering data investigation
Edge-AI RL + AHP Framework [7]	RL with multi-criteria decision making; 23.5% wait time reduction reported	Motivates RL extension; AHP approach for KPI weighting

3.2 Data Engineering Pipeline

A complete, modular data engineering pipeline has been implemented in Python across four files: `ingestion/config.py`, `ingestion/storage.py`, `ingestion/scraper.py`, and `ingestion/run_scraper.py`. The pipeline extracts 30-minute interval energy data from Newcastle University's metering portal using authenticated POST requests, with parallel execution via `ThreadPoolExecutor` and a DuckDB-backed checkpoint table enabling fault-tolerant, resumable scraping.

All project data is stored in a single portable DuckDB analytical database (`ev_twin.duckdb`). Table 2 describes the database schema.

Table 2: DuckDB database tables and their purpose.

Table	Rows	Source	Purpose
`charging_sessions`	29,775	Supervisor CSV	Primary simulation calibration data
`metering_raw`	28,728,144	NCL metering portal	Building energy data (investigated)
`simulation_params`	7 sets	Derived from EDA	Agent calibration parameters
`scrape_checkpoint`	598,596	System-generated	Fault-tolerant resume mechanism

Source code is version-controlled at: `https://github.com/NguyenIslandBoy/ev-digital-twin`

3.3 Metering Data Investigation

To assess whether building switchboard meter data (`USB_E_SwitchMA` and `USB_E_SwitchMB`) could be integrated with charging session data for grid load validation, a systematic data quality investigation was conducted. Charging sessions were aggregated to 30-minute windows and joined on timestamp across 57,696 matched intervals.

Two critical findings emerged. First, the Pearson correlation between aggregated session energy and switchboard meter readings was $r = -0.086$, indicating no meaningful relationship. Second, monthly analysis of distinct meter values per interval revealed that 34 of 40 months contained a single repeated value across all 48 daily slots — indicative of metering system collection failures rather than genuine interval consumption data. Only 6 months yielded real interval-level readings, representing less than 15% of the study period. This was confirmed by the project proposer, who verified these are the only meters capturing EV-related load in the USB building.

Decision: `usb_merged_final_data.csv` is used as the sole data source. Grid load KPIs will be derived from simulation output rather than real meter readings. The metering investigation is documented as a data quality finding demonstrating analytical rigour.

3.4 Exploratory Data Analysis

A comprehensive EDA was conducted on 29,775 charging sessions in `notebooks/01_eda.ipynb`, following established approaches for characterising empirical EV charging behaviour [15]. Key findings that directly inform agent behaviour parameters are summarised in Table 3.

Table 3: Key EDA findings and their simulation implications.

Finding	Detail	Simulation Implication
Bimodal arrival pattern	Peaks at 11:00–12:00 and 14:00–16:00, consistent with lecture schedule	Empirical hourly arrival rates; Poisson sampling per hour
Weekday vs weekend split	68.93% weekday sessions; weekend peak shifts to 10:00–11:00	Separate arrival distributions by day type
Connector type distribution	IEC_62196_T2_COMBO: 82.07%, CHADEMO: 14.32%, IEC_62196_T2: 3.61%	Categorical sampling per agent
Energy distribution	Mean 30.4 kWh; flat plateau 5–20 kWh; Gamma fit poor (AIC-selected but mixture distribution evident)	Empirical sampling per connector type
Duration distribution	Gamma fit confirmed (AIC-selected); median 0.64 hrs; capped at 4 hrs (99.9th percentile)	Gamma sampling per connector type
Charger utilisation	6 chargers; 5000198 and 5000199 handle ∼48% of sessions	Per-charger capacity modelled explicitly
Seasonal pattern	Spring highest, Summer lowest (term-time effect)	Season-aware arrival scaling in scenario engine

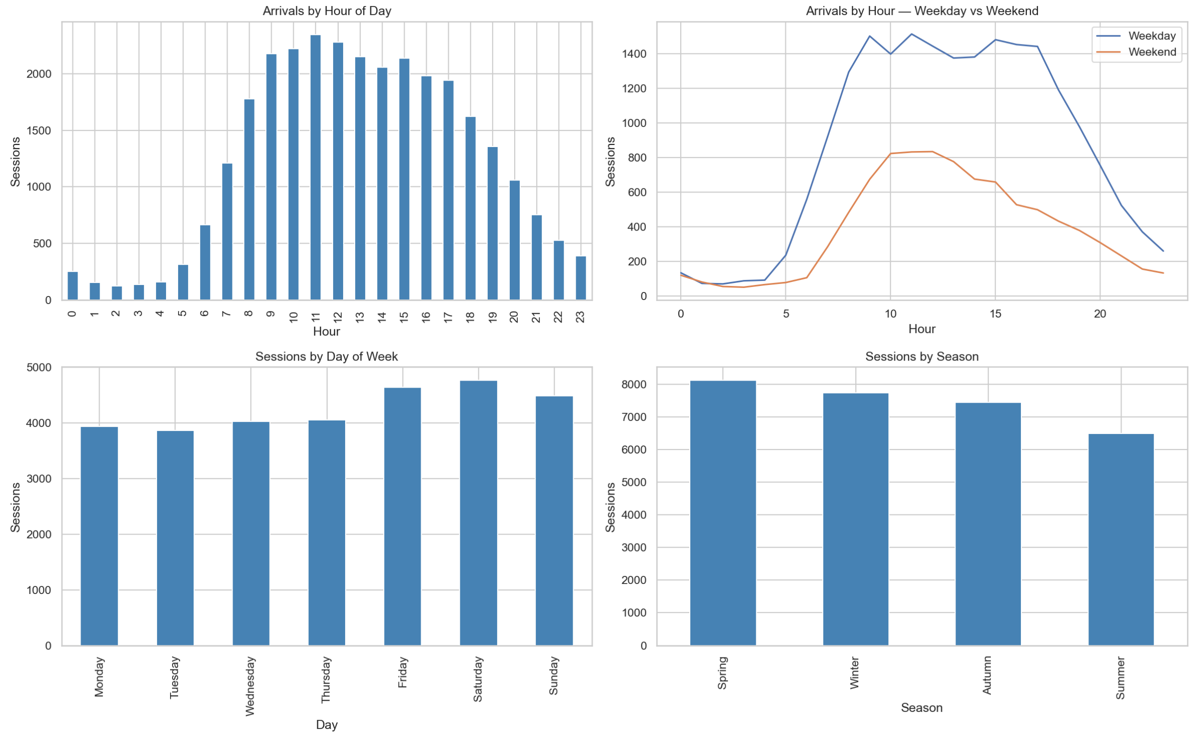

Figure 1: Hourly arrival distributions across all sessions (left), and weekday vs weekend comparison (right). Bimodal peaks at 11:00–12:00 and 14:00–16:00 are consistent with the USB building lecture schedule.

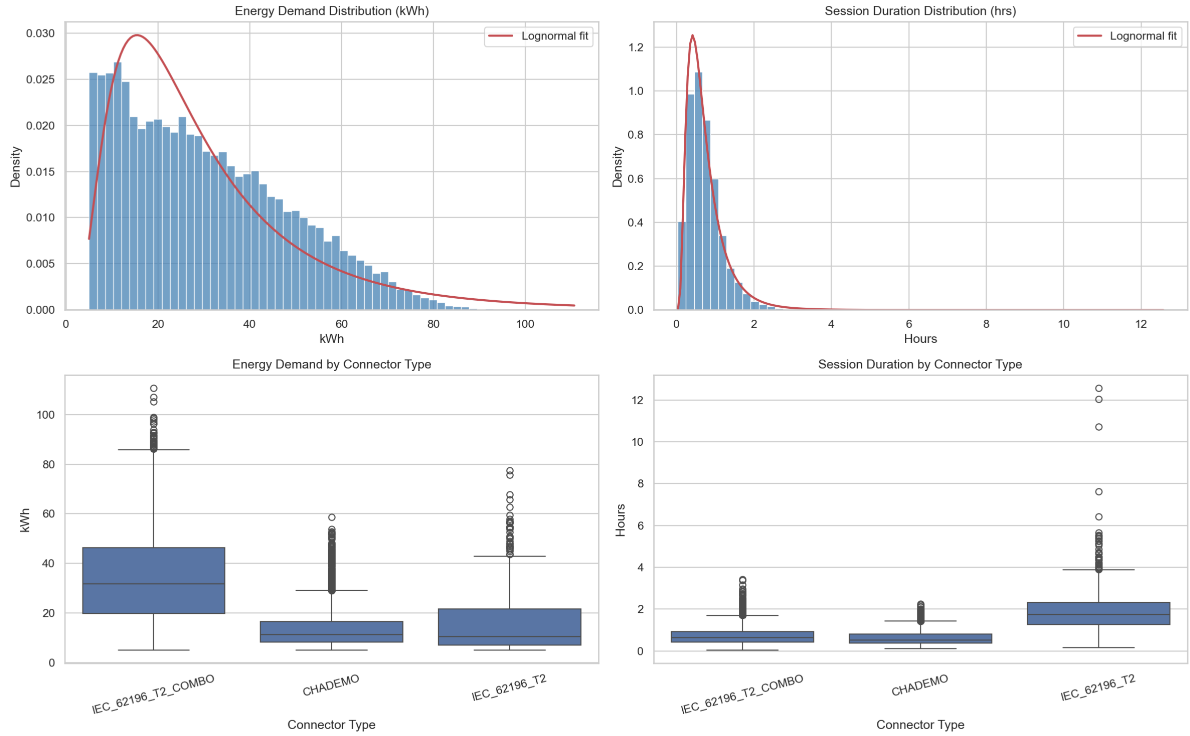

Figure 2: Energy demand and session duration distributions with Gamma fits (AIC-selected), and per-connector type boxplots. The Gamma fit is poor for energy due to a flat plateau between 5–20 kWh, motivating empirical sampling; the Gamma fit is good for duration.

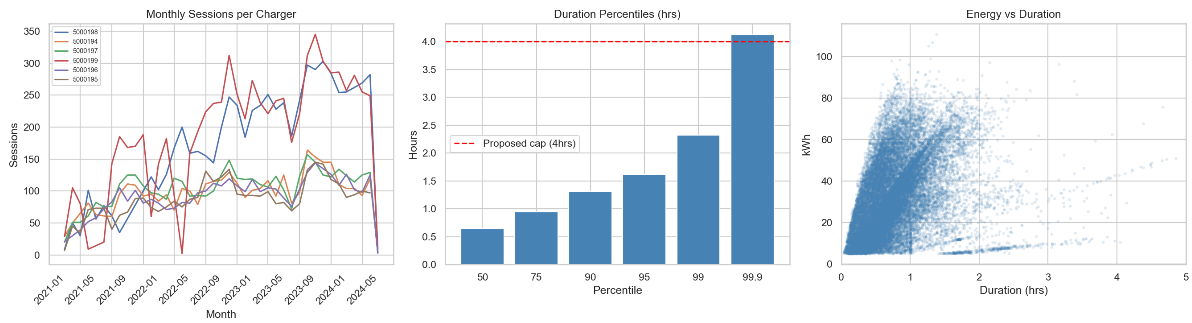

Figure 3: Monthly session counts per charger (left), duration percentile analysis with proposed 4-hour cap (centre), and energy vs duration scatter plot (right).

3.5 Key Design Decisions

Table 4 summarises the major technical decisions made and their justifications.

Table 4: Key technical design decisions and justifications.

Decision	Choice	Justification
Storage	DuckDB	Zero server setup, columnar analytics, portable, native pandas integration
Simulation framework	Mesa [8, 10]	Python-native ABM, appropriate for 6-charger network scale, reproducible
Energy demand modelling	Empirical sampling per connector type	Lognormal fit visually poor; flat plateau 5–20 kWh indicates mixture distribution
Duration modelling	Gamma per connector type (AIC-selected)	Gamma fit visually confirmed; separate params per connector captures behavioural heterogeneity
Metering integration	Not integrated	Collection failures in 34/40 months; $r = -0.086$; confirmed by supervisor
RL framework	Stable-Baselines3 PPO on Colab [9, 12]	Production-grade library; PPO stable for continuous action spaces; Colab A100 for compute

4 Project Plan

4.1 Phases and Timeline

Table 5 summarises the five project phases, linked to the objectives defined in Section 2.2.

Table 5: Project phases, tasks, and target completion dates.

Phase	Key Tasks	Objectives	Target
Phase 1: Data Engineering	Pipeline, DuckDB storage, metering investigation, EDA	O1, O2	Complete (May 2026)
Phase 2: Mesa Simulation	Agent design, charger resource model, calibration validation	O3	June 2026
Phase 3: Scenario Engine	ToU pricing, EV adoption growth, carbon incentive scenarios, KPI evaluation	O4, O5	Mid July 2026
Phase 4: RL Extension	Gym environment wrapper, PPO training on Colab, policy comparison	O6	Late July 2026
Phase 5: Writing and Submission	Dissertation, source code cleanup, documentation, poster, oral presentation	All	10–13 Aug 2026

4.2 Milestones

Table 6: Project milestones aligned with official submission deadlines.

Date	Milestone
5 June 2026	Interim report submitted
30 June 2026	Calibrated Mesa simulation validated against historical data
20 July 2026	All three what-if scenarios evaluated with KPI report
29 July 2026	Research poster submitted (10 pts)
10 August 2026	Dissertation and source code submitted
13 August 2026	Oral presentation evidence submitted

4.3 Gantt Chart

4.4 Risks

Table 7: Project risk register with likelihood, impact, and mitigation strategies.

Risk	Likelihood	Impact	Mitigation
Mesa simulation fails to calibrate within tolerance	Medium	High	Use KS-test for distribution comparison; fall back to empirical session replay if needed
RL agent fails to converge	Medium	Medium	RL is an extension; core deliverable is the scenario engine; reduce state space if needed
Dissertation writing underestimated	High	High	Budget 5 weeks minimum; begin writing in parallel with simulation development in June
Session cookie expiry during scrape	Low	Low	Checkpoint table enables resume; scrape is already complete

5 Data Management Plan

5.1 Data Description

Table 8: Description of all data assets used and generated by this project.

Data	Source	Format	Size	Sensitive?
Charging sessions	Dr Sanchari Deb	CSV	∼4 MB	No — auth IDs are anonymised integers
Building metering	NCL metering portal	DuckDB	∼400 MB	No — institutional energy readings
Simulation params	Derived from EDA	JSON	<1 MB	No
Python source code	Student-created	.py files	<1 MB	No
Jupyter notebooks	Student-created	.ipynb	<5 MB	No
RL model weights	Student-created (Colab)	.zip (SB3)	<50 MB	No

5.2 Data Collection and Quality

The charging session dataset was provided directly by the project proposer and is treated as the authoritative source. No modifications are made to the raw CSV file; all transformations are applied at load time via DuckDB SQL and documented in the project README and EDA notebook.

Building metering data was collected via an automated Python scraper using authenticated POST requests to the university metering portal. A checkpoint table in DuckDB records the status of every (meter_id, date) pair scraped, providing full audit coverage of 598,596 requests. Data quality issues identified during the metering investigation are documented in the EDA notebook with supporting SQL queries and correlation analysis.

5.3 Storage and Backup

All code and notebooks are version-controlled at `https://github.com/NguyenIslandBoy/ev-digital-twin`. Raw data and the DuckDB database are excluded via `.gitignore`. Table 9 summarises storage locations.

Table 9: Storage and backup strategy for all project assets.

Asset	Primary Location	Backup
Source code and notebooks	GitHub (private)	Local machine
Raw CSV dataset	Local machine (`data/raw/`)	University OneDrive
DuckDB database	Local machine (`data/`)	External drive (weekly)
Trained model weights	Google Colab / Drive	Local machine after training

5.4 Ethics, Privacy, and Legal Compliance

This project does not involve human participants, surveys, interviews, or the collection of personal data. Charging session auth IDs are anonymised integer identifiers with no linkage to named individuals. The dataset was provided by the project proposer under institutional data sharing, and building metering data contains no personal identifiers.

Ethical approval has been sought via the Newcastle University ethics portal as required by CSC8639. The project is classified as low-risk: no personal data, no human participants, no data collected outside the UK or EEA, and no commercially sensitive information. Newcastle University's Research Data Management Policy and Information Security Policy are adhered to throughout.

5.5 Reproducibility and Sharing

The project is designed for full reproducibility from raw data to final outputs. A `requirements.txt` and `pyproject.toml` document all Python dependencies; a `README.md` documents environment setup and execution steps; and all notebooks are structured to run sequentially from DuckDB to final plots. After dissertation submission, the GitHub repository will be made public. Raw charging session data will not be shared publicly without supervisor approval.

5.6 Retention and Deletion

All project data will be retained until dissertation assessment is confirmed complete, in accordance with Newcastle University's Research Data Management Policy. No personal data requiring scheduled deletion is held. Local copies of the database and raw data will be retained for a minimum of 12 months post-submission.

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